

Basic Concepts of Biotechnology

Biotechnology is the application of **scientific principles** for **human welfare** using biological systems. To understand this field, we must first understand the basic characteristics of life.

Characteristics of Life

Life is difficult to define philosophically, but in science, it is understood through four primary characteristics:

1. **Reproduction:** The primary factor that differentiates living from non-living things. It is essentially **cell division**.
2. **Cell Structure:** Every living organism is made of cells. **No cell means no life**.
3. **Life Processes (Metabolism):** The sum total of all **bio-chemical reactions** occurring in the body or a cell.
4. **Adaptation:** The ability to adjust to a changing environment, which can lead to **Evolution** over the long term.

Key Point: Metabolism provides the **matter** and **energy** that allow a cell to grow and eventually divide (reproduce).

Metabolism: The Chemical Engine of Life

Metabolism consists of two types of reactions:

- **Catabolism:** A **breakdown** reaction where matter is destroyed to release **energy** (e.g., digestion).
 - **Anabolism:** A **synthesis** or constructive reaction where energy is used to create **matter**.
-

Viruses: The "Terrorists" of Biology

Viruses are unique because they are the **connecting link** between living and non-living things.

Why Viruses are Non-Living:

- They have **no cell structure** (no "lab" for chemical reactions).
- They have **no independent metabolism**.

Why Viruses are Living:

- They contain genetic material (**DNA or RNA**).
- They can **reproduce**, but only by hijacking a host cell.

Viral Functioning

1. **Entry:** The virus enters a living cell.
 2. **Hijacking:** It takes over the cell's cellular machinery and metabolism.
 3. **Replication:** The cell is forced to make viral proteins (matter) using its own energy.
 4. **Destruction:** The cell eventually dies and releases new viruses to infect other cells.
-

DNA and RNA: The Instruction Manuals

Cell Components for Protein Synthesis

- **Nucleus:** The "Brain" or "Control Room" of the cell; it contains the DNA.
- **DNA (Deoxyribonucleic Acid):** The **instruction manual** or blueprint of the cell.
- **RNA (Ribonucleic Acid):** The **messenger** that carries instructions from the DNA to the rest of the cell.
- **Ribosomes:** The "machines" that read the RNA and **synthesize protein**.
- **Mitochondria:** The **Powerhouse** of the cell that provides energy in the form of **ATP**.

The Central Dogma of Biology

This is the main principle of how life functions:

DNA → RNA → Protein.

1. **Transcription:** DNA transfers information to RNA (occurs in the **nucleus**).
2. **Translation:** RNA and ribosomes work together to make protein (occurs in the **cytoplasm**).

Remember: In humans, DNA is the genetic material. However, in **Retroviruses**, RNA can sometimes produce DNA through **Reverse Transcription**.

Genomics and the Human Genome

- **Gene:** A segment of DNA that holds information for a specific character. Humans have approximately **30,000 genes**.
- **Genome:** The complete set of all genes in an organism.
- **Genomics:** The study of the entire genome.
- **Transcriptome:** All the RNA molecules in a cell (the "active" genes).
- **Proteome:** All the proteins in the body.

Indian Genomics Projects

1. **IndiGen Project (2019):** Led by **CSIR**, it sequenced 1,000 Indian genomes to understand the genetic diversity of the Indian population.
2. **Genome India Project (GIP, 2020):** Led by the **Department of Biotechnology (DBT)** and **IISc Bangalore** to create a larger grid of Indian genetic data.

Key Point: India needs its own genomic data because western data (from the US/Europe)

does not accurately represent Indian risks for diseases like **diabetes** and specific **cancers**.

Ethical and Social Concerns

The use of genomic data brings several risks:

- **Discrimination:** It may create bias in employment or insurance sectors based on genetic risk.
- **Privacy:** Genomic data is highly personal and can reveal sensitive information about lineage and ancestors.
- **Social Impact:** It could shatter existing belief systems or social fabrics.
- **Cost:** High costs and the need for advanced infrastructure are significant barriers.

Advanced Genomic Terminology

1. Dark Genome

- **Definition:** The parts of the DNA that are **non-active** or considered "**junk**" because they do not code for proteins.
- **Key Statistic:** In the human body, only **2% of genes are active**. The remaining **98%** constitutes the Dark Genome.
- **Significance:** Recent research suggests that the Dark Genome may be linked to complex conditions like **Schizophrenia** and **Bipolar Disorder**.

2. Personal Genomics

- **Concept:** Sequencing the genome of a specific individual to understand their unique genetic makeup.
- **Cost:** Currently costs approximately **₹1 Lakh**.
- **Benefits:**

1. **Personalized Medicine:**
Tailoring medical treatment to an individual's genetic profile.
2. **Risk Assessment:** Identifying the likelihood of developing diseases like **Cancer** (e.g., checking for the BRCA mutation).

3. Metagenomics

- **Concept:** The study of genetic material recovered directly from **environmental samples**.
- **Process:** Instead of studying one organism, you take a sample (like soil or water), separate all the DNA from the microorganisms present, and study them **collectively**.
- **Sub-fields:** Includes Soil, Marine, and Human Gut metagenomics.

4. Environmental DNA (eDNA)

- **Definition:** DNA that is collected from the environment (water, air, soil) rather than directly from a living organ.
- **Sources:** Secreted through feces, mucus, gametes, skin, or hair.
- **Usage:** Used in **Biodiversity Conservation** to track animal migration and presence without needing to see the physical animal.

Key Point: The difference between Metagenomics and eDNA is that Metagenomics focuses specifically on **microorganisms**, while eDNA includes DNA from **all living beings** (animals, birds, etc.) in that environment.

5. Cell-Free DNA (cfDNA)

- **Concept:** Fragments of DNA that circulate freely in the bloodstream, outside of a cell.

- **Origin:** When a cell dies, its nucleus and mitochondria break down, releasing DNA into the body fluids.
- **Applications:**
 - **Gemini Test:** Used for non-invasive cancer diagnosis (e.g., Lung Cancer).
 - **NIPT (Non-Invasive Prenatal Testing):** Screening a fetus for genetic abnormalities (like Down Syndrome) by testing the mother's blood, which contains the baby's cfDNA.

Remember: NIPT is safer than **Amniocentesis** because it doesn't require a needle to enter the womb, eliminating the risk of miscarriage.

Genetic Engineering: The Case of Insulin

Genetic engineering is the deliberate modification of an organism's characteristics by manipulating its genetic material.

Process of Producing Human Insulin:

1. **Isolation:** Identify the insulin-producing gene in the **Beta cells** of the human Pancreas.
2. **Cutting:** Use **Restriction Endonuclease (RE)** (known as "Molecular Scissors") to cut the specific gene.
3. **Vector Preparation:** Take a **Plasmid** (extra-chromosomal DNA) from an *E. coli* bacteria and open it using the same RE enzyme.
4. **Ligation:** Insert the human insulin gene into the bacterial plasmid using **Ligase** (known as "Molecular Glue").
5. **Transformation & Multiplication:** Put the modified plasmid back into the

bacteria. As the bacteria multiplies, it produces human insulin.

Genetically Modified (GM) Crops

Crops that have been modified using genetic engineering to improve yield, nutrition, or resistance.

- **Golden Rice:** Modified to contain high levels of **Vitamin A**.
 - **GM Rubber:** Developed to survive in cold temperatures (pioneered in Assam, India).
 - **DMH-11 (Dhara Mustard Hybrid):** A GM mustard variety designed to increase yield by 25-30%.
 - **Types of Modification:** * **Cisgenic:** Moving genes between the same or closely related species.
 - **Transgenic:** Moving genes between different species.
-

Stem Cells – Complete Notes

What is a Stem Cell?

- Stem cells are non-specialised / undifferentiated cells present in animals.
- They can divide and produce different types of specialised cells in the body (muscle cells, blood cells, skin cells, bone cells, etc.).
- Stem cells themselves have no specific function in the body except to divide.
- Once they divide, the new cells become specialised / differentiated and perform specific jobs.

Analogy: Just like a plant stem can give rise to leaves, flowers, fruits and branches, stem cells can give rise to any required cell type.

Location:

- Mainly in bone marrow (soft jelly-like tissue inside hollow long bones, e.g., femur).
- Also present in other places in the body.

Blood formation:

- RBC, WBC and platelets are formed from stem cells present in bone marrow.
-

Three Main Characteristics of Stem Cells

1. **Self-renewal capacity** – They can renew (make copies of) themselves.
2. **Multiple division cycles** – They can divide many times.
3. **Differentiation** – After division, they can produce different types of specialised cells.

Key Point: Stem cells act as an internal repair system – they supply new cells for growth, repair and replacement of dead cells.

Functions of Stem Cells in Our Body

- Supply new cells for embryonic development and growth after birth.
 - Supply cells for repair and healing (e.g., wound healing after cut or scratch).
 - Replace dead cells in the body (e.g., RBCs live only 120 days; new RBCs are made by stem cells in bone marrow).
-

Types of Stem Cells (Based on Potency)

Type	Full Form	Potential (What they can form)	Present in	Medical Use
Totipotent	Total potential	All 500+ cell types in the body	Early embryonic stage (1-week embryo)	No medical use (globally banned)
Pluripotent	Many potential	Majority of cell types	Later embryonic stage	Most important for therapy
Multipotent	Multiple potential	Many (100–150) related cell types	Adult tissues	Limited use
Oligopotential	Few potential	Few cell types	Adult tissues	Very limited
Unipotent	One potential	Only one specific cell type	Adult tissues	Very limited

Remember:

- Totipotent & Pluripotent → mainly embryonic stem cells.
- Multipotent, Oligopotential & Unipotent → mainly adult stem cells.
- Potency decreases as we move from Totipotent → Unipotent, but specialisation increases.

Totipotent Stem Cells

- Can form any cell type in the body.
- Present only in 1-week-old embryo.
- No medical use – globally banned.
- Reason: Retrieving them destroys the embryo (ethical & legal issue – foeticide).

For research only: Totipotent cells can be obtained from embryos developed by IVF (In Vitro Fertilisation) in the lab (not from mother's womb).

Pluripotent Stem Cells (Most Important)

- Can form majority of cell types in the body.
- Developed by division of totipotent cells.
- Present in later embryonic stage (after 1 week, up to 9 months).
- Medically most useful for stem cell therapy.
- Embryo is not destroyed if retrieved carefully.

How to retrieve Pluripotent Stem Cells?

1. From later embryonic stage (technically possible but banned – embryo may die).
2. From miscarriage (within 12 weeks) – only for research.
3. From umbilical cord blood at the time of birth (most popular & safe method).

Key Point: Umbilical cord belongs to the baby, not the mother. After birth, cord is cut anyway → cord blood (containing pluripotent stem cells) can be collected safely.

Stem Cell Preservation & Banking

- Pluripotent stem cells from umbilical cord blood are preserved.
- Preserved in liquid nitrogen at -196°C (cryopreservation).
- Also known as Cord Blood Banking.

Types of Stem Cell Banking in India

1. Private Banking

- 100% ownership to parents.
- No one can use without permission.

2. Public Banking

- Donated stem cells.
- Anyone can use by paying service fee.

3. Community Banking

- Done in a group/community.
- Anyone within the community can use.

Usage:

- Stem cells are mainly used for the same person from whom they were taken.
- In many cases, they can also be used for close relatives (siblings, parents, offspring) after proper matching (blood group + HLA testing).

Induced Pluripotent Stem Cells (iPS Cells)

- Developed by Dr. Shinya Yamanaka (Japan) in 2007.
- Nobel Prize in Medicine 2012.
- Normal body cell (e.g., skin cell) is taken → gene reprogramming → converted into pluripotent stem cell.

- Useful for people whose cord blood was not preserved.

Stem Cell Therapy (Applications)

Pluripotent stem cells can be used for treatment of:

1. Blood-related diseases:

- Leukaemia (blood cancer)
- Anaemia (low haemoglobin)
- Sickle cell anaemia
- Thalassemia

2. Tissue damage due to severe burns and wounds.

3. Neurodegenerative diseases:

- Alzheimer's disease (memory loss)
- Parkinson's disease (loss of body control)

4. Eye disorders – Construction of cornea and retina to treat blindness.

5. Diabetes – Generate insulin-producing cells.

6. New organs and tissues (future possibility) for organ transplant.

7. Spinal cord injury, heart disease, liver disease, etc.

8. Baldness (hair regrowth).

Note: Success depends on doctor's advice. Not all diseases can be treated yet.

Stem Cell Research in India

Milestones:

- 2002: Dr. Venugopal (AIIMS Delhi) developed stem cell micro-injection for heart disease treatment.

- **2005: India's first embryonic stem cell bank for research established at AIIMS Delhi.**
- **2005: Centre of Excellence for Stem Cell Research established at CCMB Hyderabad.**
- **2007: Government released National Stem Cell Policy to regulate commercial stem cell banking.**
- **Stem cell banking regulated by DBT and ICMR.**
- **2019: Stem cells included under New Drugs and Clinical Trials Rules 2019 → stricter government control.**
- **All hospitals/institutions doing stem cell research must have an Institutional Committee registered with National Apex Committee for Stem Cell Research and Therapy (NAC-SCRT).**

3. Organism cloning (reproductive cloning)

Therapeutic vs Reproductive Cloning:

- **Therapeutic cloning: Used to produce cells/tissues/organs for treatment (already discussed in stem cell therapy).**
- **Reproductive cloning: Produces entire new organism (e.g., Dolly the sheep).**

Famous Examples:

- **1996: First animal clone – Dolly the sheep (Scotland) by Ian Wilmut using SCNT (Somatic Cell Nuclear Transfer) method.**
- **2009: India's first buffalo clone – Garima (NDRI Karnal) using Hand-guided Cloning Technology (modified SCNT).**
- **2012: First Pashmina goat clone – Noori (NDRI + Sher-e-Kashmir University).**
- **2014: Clone of critically endangered wild buffalo of Chhattisgarh – Deepasha.**
- **2015: Buffalo clone Swrupa with high milk yield (25.1 L/day).**
- **2023: First Gir cow clone – Ganga (15 L milk/day).**

Purpose of Cloning in India:

- **Enhance milk and meat production in cattle.**
- **Increase production of commercially important products (e.g., Pashmina wool).**
- **Promote stem cell-based regenerative medicine and hormone production.**
- **Conserve critically endangered animal species.**

Cloning

Definition:

- **Cloning is the process of making genetic copies of an organism or any biological entity (gene, cell, tissue, organ, or organism).**
- **Clones are produced by asexual reproduction (single parent involved).**
- **Clones have same genetic, physical and biological characters as the parent (but psychological/acquired characters may differ).**

Types of Cloning:

- 1. Gene cloning**
- 2. Cell, tissue & organ cloning (includes therapeutic cloning)**

Issues with Cloning:

- **Low success rate (~10%).**
 - **Clones inherit older DNA → higher risk of diseases and early ageing (concept of genetic age).**
 - **Extensive use of cloned animals → loss of local breeds and biodiversity.**
 - **Ethical concerns – interference with natural biological systems.**
-

Human Cloning

- **Technically possible but globally banned.**
 - **Banned due to socio-ethical concerns:**
 - **Issues related to family rights, social rights and citizen rights.**
 - **Problems in relationship of clone with family members.**
 - **Promotion of single parenthood.**
 - **Misuse of human clones as medical resource (organ harvesting), frontline human shield in war/terrorism.**
 - **In India: ICMR guidelines prohibit cloning of human embryo.**
 - **Animal cloning is permitted for research and agriculture use.**
-

Legal Status of Cloning in India

- **Human cloning – Prohibited under ICMR guidelines and bioethics policies.**
- **Animal cloning – Allowed for research and agricultural purposes.**

1. Altruistic Surrogacy in India

Eligibility Criteria for Couples Availing Surrogacy

- **Only Indian citizens, NRIs, and OCIs are allowed.**
 - **OCI full form:** Overseas Citizen of India.
- **Must be heterosexual married couples (male + female).**
- **Married for at least 5 years.**
- **Childless – No surviving child (biological or adopted).**
 - One child is considered enough; surrogacy not allowed if any surviving child exists.
- **Age range:**
 - Wife (female): **23–50 years**
 - Husband (male): **26–55 years**

Remember: Purpose is to stop misuse of surrogacy.

Who cannot avail surrogacy?

- Single males (any category)
- Unmarried single females (except special cases)
- Live-in couples
- Homosexual couples
- Foreigners (completely banned)

Exception for single females:

- **Divorced or widow females** can avail surrogacy (sperm from anonymous donor).
- **Vikky Donor** movie example given for anonymous sperm donor.

Key Point: Only close relatives can be surrogate mothers – no commercial surrogacy.

Eligibility Criteria for Surrogate Mother

- Must be a **close relative** of the intending couple.
- Must be a **married female**.
- Must have **at least one biological child** of her own.
- Must be **physically and mentally fit**.
- **Age bracket: 25–35 years**.
- Can provide **only one surrogacy** in her entire lifetime.
- **Her own egg and her husband's sperm cannot be used** in the surrogacy (to avoid biological complications and traditional surrogacy).

Financial transaction not allowed – purely altruistic (no payment to surrogate).

Regulation of Altruistic Surrogacy in India

- Regulated under **Surrogacy (Regulation) Act, 2021** (often referred as 202).
- **National and State Surrogacy Boards** established for regulation and control.
- **Punishment** for breach of law:
 - Up to **10 years** imprisonment.
 - Penalty up to **₹1 lakh**.

Process:

1. Intending couple finds eligible close relative surrogate (25–35 years, all criteria fulfilled).
2. Apply to Surrogacy Regulation Board.
3. Board reviews and approves only if all conditions met.

2024 Amendment (Important Update)

- **Donor sperm or donor egg** now allowed **if one partner has medical**

condition and cannot produce sperm/egg.

- Earlier: Complete ban on donors (only couple's own gametes).
- Now: Relaxation for medical reasons.

Note: This makes the process slightly easier for couples with infertility issues in one partner.

Issues & Controversies Related to Surrogacy Law

- **Limiting in nature** → Scope of surrogacy in India reduced by **99%**.
 - Social stigma: Asking sister/sister-in-law for surrogacy is socially awkward/messy.
- **Gender bias:**
 - Divorced/widow females allowed.
 - But **divorced/widower/single males** not allowed.
 - **Purush Mukti Morcha** (Men's Rights group) advocates for single/divorced/widower males' rights.
- **LGBTQ+ community ignored:**
 - No provision for homosexuals, live-in couples, or transgenders.
 - For transgenders, surrogacy is often the **only option** to have children.
 - Society wants LGBTQ+ in mainstream (jobs, education) but law does not support their right to family via surrogacy.

Remember: Law is very restrictive; social acceptance is low; gender and LGBTQ+ rights issues exist.

2. DNA Profiling / DNA Fingerprinting

What is DNA Profiling / DNA Fingerprinting?

- Technique to **establish identity** of a person or organism.
- Based on **differences and similarities** in DNA sequences.
- No two individuals (except identical clones) have **100% same DNA**.
 - Humans are **99.9% same, 0.1% different**.
 - Close relatives have more similarity; unrelated people have more differences.
- Used to **zero in on criminals**, identify victims, determine paternity, etc.

Applications (Forensic + Others):

- Crime scene & criminal investigation
- Identifying culprits
- **Paternity/maternity cases** (biological relationship)
- Identification of **dead bodies in mass disasters** (e.g., Rajiv Gandhi assassination bomb blast; Ahmedabad plane crash where bodies were charred at 1000°C due to 1 lakh litres gasoline)
- Tracing & identifying **missing children and adults**
- Medical research – identifying **genetic diseases**
- **Wildlife protection** – tracing poached animals to specific forests/jungles by matching DNA with local animal families (territorial animals like tigers in Sundarbans vs Jim Corbett have unique genetic signatures)

DNA Barcoding (Related concept):

- Used to identify **different plant and animal species**.
- Mainly for **wildlife & biodiversity studies** + wildlife forensic studies.
- Difference: DNA profiling = individual level; DNA barcoding = species level.

How is DNA Profiling Done? (Technology Used)

Key Technologies:

1. PCR (Polymerase Chain Reaction)

- Used to **increase (amplify) the amount of DNA** collected from crime scene (often very small quantity – blood spot, skin cells, etc.).

2. Southern Blotting Technology (SBT)

- Identifies **specific DNA sequences** (VNTRs & STRs) in the entire DNA content.

3. Gel Electrophoresis

- Separates DNA fragments (VNTRs & STRs) **on the basis of size and charge**.
- DNA is **negatively charged** → moves towards positive terminal.
- Heavier fragments move less distance; lighter move farther.

VNTR & STR (Important Terms):

- **Variable Number of Tandem Repeats (VNTR)** and **Short Tandem Repeats (STR)** / Microsatellite DNA.
- These specific repeated sequences are present in **every human**.
- **Number and location** differ between individuals → creates unique “fingerprint”.

Process Flow:

1. Collect tiny DNA sample from crime scene.
2. Amplify using PCR.
3. Identify VNTR/STR using Southern Blotting.
4. Separate fragments using Gel Electrophoresis.
5. Compare band patterns with suspects/victims.

Samples collected from:

- Blood, sperm, **root hair** (not cut hair), skin cells, saliva, etc.

Development:

- Developed by **Sir Alec Jeffreys (UK)** in **1984**.
- In India: Developed in **1988** at **CCMB Hyderabad** (Centre for Cellular & Molecular Biology).
- For judicial purposes: Done only at **CDFD Hyderabad** (Centre for DNA Fingerprinting and Diagnostics).

Other Related Blotting Techniques (Easy to remember by directions):

- **Southern Blotting** → DNA (named after scientist Southern)
- **Northern Blotting** → RNA
- **Western Blotting** → Protein (confirmatory test for HIV after ELISA)
- **Eastern Blotting** → Protein modifications

ELISA:

- Easy & quick primary test for HIV.
- If positive → confirm with Western Blotting (more accurate, time-consuming).

Gel Electrophoresis also works for RNA and protein.

Forensic Biotechnology (Related Techniques)

- DNA Profiling
- Polygraphy (Lie detector test)
- Narco Test (Truth serum test)
- Brain Mapping / Brain Fingerprinting

3. CRISPR-Cas9 (Gene Editing Tool)

Discovery & Nobel Prize:

- Discovered in **1987** in bacteria (natural defence system).
- First used for human genome editing in **2012**.
- **Nobel Prize in Chemistry 2020** awarded to **Emmanuelle Charpentier** and **Jennifer A. Doudna**.

Natural Function (Bacteria vs Bacteriophage):

- Bacteria attacked by bacteriophage (virus).
- Virus inserts its DNA (like AK-47 + RDX) into bacteria.
- Bacteria's **CRISPR** segment (part of its DNA) transcribes into **Guide RNA (gRNA)**.
- gRNA has complementary sequence to viral DNA.
- **Cas9 enzyme** attaches to gRNA.
- Complex finds and **cuts** viral DNA → infection destroyed.

Mechanism (Step-by-Step):

1. CRISPR (DNA segment) → transcribed into **Guide RNA**.

2. Guide RNA has **complementary sequence** to target DNA.
3. Guide RNA + **Cas9 enzyme** form complex.
4. Complex binds to target DNA by sequence complementarity.
5. Cas9 cuts the target (faulty) gene from genome.

Benefits / Applications:

- **Gene therapy & human genome editing.**
- Treatment of **cancer**.
- Genome editing of **plants, crops & animals**:
 - **Kamla rice (DRR Dhan 100)** – genome edited.
 - **Pusa DST rice** – genome edited variety.
 - India's first genome-edited sheep (by Sher-e-Kashmir University) – muscle mass increased by 30%.
- Edit genome of **microorganisms** (bacteria & fungi) for better **industrial applications** (biofuel, biochemicals production – bioethanol, biodiesel, biogas, enzymes, pigments).
- Achieve **immunity against viral diseases** (COVID-19, HIV, others).

Remember: CRISPR-Cas9 is one of the most accurate & advanced genome editing tools.

- **T-cell** = Type of WBC (lymphocyte) – killer cell.
- **Therapy** = Gene therapy using CRISPR-Cas9.

Why needed?

- Cancer cells are our own cells → have normal antigens + special **chimeric antigens**.
- Normal T-cells cannot recognize cancer cells (self-cells).
- CAR-T modifies T-cells to recognize chimeric antigens.

Process (Step-by-Step):

1. Blood taken from cancer patient → T-cells harvested.
2. In lab: Gene for CAR inserted into T-cells using **CRISPR-Cas9**.
3. Modified T-cells (now CAR-T cells) produce **Chimeric Antigen Receptor** protein on surface.
4. CAR-T cells multiplied into millions in lab.
5. Infused back into patient.
6. CAR-T cells bind to cancer cells (via chimeric antigen) and kill them.

Indian Development (2023):

- **NexCAR19** – India's first indigenous CAR-T cell therapy.
- Developed jointly by **IIT Bombay, Tata Memorial Centre & ImmunoACT**.
- Approved for patients **aged 15 and above**.
- Especially for **blood cancers** when other treatments fail.
- **Success rate:** ~70%.

4. CAR-T Cell Therapy (Cell-Based Gene Therapy)

Word Dissection:

- **CAR** = Chimeric Antigen Receptor (synthetic receptor).

- Cost reduced from crores (foreign) to **20–40 lakhs**.

Side Effects:

- Neurotoxicity (affects nervous system) – needs management.
- It is **immunotherapy**, not chemotherapy.

Type: Example of **somatic cell gene therapy / cell-based gene therapy**.

5. Gene Silencing

Definition:

- Technique to **block expression of a gene** (protein synthesis stopped) without changing the gene itself.
- Target: **Messenger RNA (mRNA)** destroyed → no harmful protein formed.

RNA Interference (RNAi)

- **Double-stranded RNA** molecule made in lab and inserted into target cell.
- This dsRNA destroys mRNA coming from faulty gene.
- Protein synthesis stopped.

Applications:

- Treatment of **HIV infection**.
- Lower **cholesterol level** in blood (prevents cardiovascular diseases).
- Developing **anti-viral defence mechanisms** in humans → helpful in managing pandemics like COVID-19.

Another Example:

- **Antisense technology** (older, less popular).

Examples of Gene Therapy:

- Sickle cell anaemia

- Haemophilia
- NexCAR19 for blood cancer

Regulation in India

- Regulated by **ICMR (Indian Council of Medical Research)** and **DBT (Department of Biotechnology)**.
- Comprehensive **National Guidelines** released in **2019**.
- Ensure **controlled clinical trials, safety & ethical oversight**.

Current Status:

- India ranks among **top Asian countries** in gene therapy research.
- Indian gene & cell therapy market valued at **over \$750 million**.

Challenges:

- Limited companies active.
- Regulatory and ethical concerns (designer babies, etc.).

1. Basic Concepts of Nanotechnology

What is “Nano”?

- **Scientific meaning:** 1 nano = 10^{-9} (one billionth).
- Example: 1 nanometre (nm) = 10^{-9} metre.

What is Nano Scale?

- Size range: **1 nm to 100 nm**.
- Anything falling in this range is studied under **nano science & nanotechnology**.
- **Macro scale** = anything above milli scale (visible to naked eye).
- **Micro scale** = biological scale (cells, bacteria) → needs microscope.

- **Nano scale** = extremely small; needs advanced tools.

Key Point: When any material is brought down to **nano scale**, it often shows **new and different properties** that are not seen at macro or micro scale.

Why New Properties Appear at Nano Scale? (Principles)

1. Quantum Physics Dominates

- At nano scale, behaviour is governed by **laws of quantum physics** (different from classical physics).

2. Extremely High Surface-to-Volume Ratio

- Surface area increases dramatically.
- Leads to:
 - Changed **physical properties** (e.g., gold becomes liquid at room temperature at nano scale).
 - Increased **chemical reactivity**.

3. Energy Exchange with Environment

- High surface area affects heat/energy exchange → alters material behaviour.

Remember: Same element can show completely different properties at different scales (e.g., silver at macro → jewellery/utensils; at micro → best conductor; at nano → kills bacteria).

Methods / Approaches of Nanotechnology

1. Top-Down Approach

- Start with bulk material (macro scale) → break it down to nano scale.
- Used for **simple nano structures**.
- **Disadvantages:**
 - Time-consuming and expensive.
 - Not suitable for **mass production** of complex nano structures (e.g., nano machines/pumps).

2. Bottom-Up Approach (Future-oriented)

- Start with atoms/molecules → assemble them into nano structures.
- Based on **chemical synthesis and molecular assembly**.
- **Advantages:**
 - Suitable for **mass production of complex nano structures**.
 - Less time-consuming (comparatively).
 - More affordable in future.

Example: Nano pump (to regulate blood flow at blockage sites) is made by bottom-up (assembling ~200 atoms/molecules).

Key Point: Future of nanotechnology lies in **bottom-up** approach.

Nano Materials

Two Main Types

1. **Non-Carbon Based**
2. **Carbon Based**

Important Examples

Non-Carbon Based:

- **Silver nano crystals** → kill bacteria (used in burn creams, wound healing).

- **Aluminium silicate** → scratch-resistant coatings (spectacles, mobile screens).
- **Zinc oxide & Titanium dioxide** → UV protection in sunscreens/cosmetics.
- **Platinum nano particles** → catalyst in fuel cells.

Carbon Based:

- **Graphene**
- **Carbon Nano Tubes (CNT)**
- **Fullerenes**

Already Available in Market:

- Wrinkle-free fabrics
- Water-resistant fabrics
- Odour-free fabrics

- **Graphene** → lighter, thinner, flexible touch-screen displays (foldable/rollable screens).
- **Graphene-based transistors** → 4 times faster than silicon transistors (IBM development).
- **Carbon Nano Tubes (CNT)** → replace wires → more energy-efficient circuits.

6. Textiles (Technical Textiles)

- Wrinkle-free, water-resistant, odour-free fabrics.
- **Self-cleaning fabrics** (using Titanium dioxide) → clean themselves in sunlight.
- Fire-resistant & protective clothing for firefighters and armed forces.
- Antimicrobial fabrics (silver nano crystals).

Major Applications of Nanotechnology

1. Health Sector

- Targeted drug delivery
- Nano medicines
- Tissue engineering
- Cancer treatment, antimicrobial coatings, etc.

2. Environment

- Water purification & filtration
- Nano sensors for pollution detection

3. Energy

- Better batteries, fuel cells, solar cells (using graphene, CNT, etc.)

4. Agriculture

- Nano fertilisers, nano pesticides, smart delivery systems

5. Electronics

7. Defence

- Bullet-proof jackets, helmets, uniforms with nano materials → lighter yet stronger.
- Nano sensors for biological attack detection in water supply/public places.

Note: Many applications are already in market (wrinkle-free shirts, scratch-resistant glasses, antimicrobial creams, water-resistant denim, etc.).

Issues & Concerns

1. Health-Related Issues

- Nano medicines/particles can cross **blood-brain barrier**.
- May interfere with brain functioning.
- Can cause side effects in other organs.
- Long-term toxicity still not fully understood.

2. Socio-Ethical Issues

- Advanced nano medicines & tissue engineering will be **very costly**.
- Only rich people will afford them → creates **social disparity**.
- May lead to “designer humans” (stronger, smarter, better-looking) → against ethics & social justice.

3. Environmental Issues

- Widespread use may increase **nano pollution** in environment.
- Concept of “**Grey Goo**” and “**Green Goo**” (from Eric Drexler’s book *Engines of Creation*):
 - **Grey Goo**: Hypothetical nano machines that destroy all physical structures.
 - **Green Goo**: Hypothetical nano toxins that destroy all life forms.

4. Economic Issues

- Development requires huge energy input (making/breaking bonds at atomic level).
- Many nano structures need rare/expensive elements (platinum, gold, titanium, etc.) → high production cost.
- Gap between laboratory research and market application → limited commercial success so far.

Development of Nanotechnology in India

Key Milestones

- **2001**: Nano Science & Technology Initiative (NSTI) launched by Department of Science & Technology (DST) → focused on infrastructure &

human resource development. 19 Centres of Excellence established.

- **2007: Nano Mission** (National Nano Mission) launched → still ongoing.
 - Objectives:
 1. Boost fundamental research in nano technology.
 2. Develop advanced infrastructure & ecosystem.
 3. Develop trained human resources.
 4. Encourage private sector participation.
 5. Promote international cooperation.
- **Nano Mission Council** established under leadership of **Prof. C.N.R. Rao** (Bharat Ratna scientist).
- **Institute of Nano Science & Technology (INST)** established in Mohali, Punjab (dedicated institute for nano science & technology).
- **India’s First Nano Park** being developed in Bengaluru (special zone for private companies working in nano technology with easy lease & infrastructure).

International Cooperation

- **ICONSAT** (International Conference on Nano Science & Technology) organised every 2 years in India.
 - Attracts ~1000 delegates from ~15 countries (researchers, academicians, industry leaders, policymakers).

Scientists & Technocrats of Indian Origin (STIOs)

- Government regularly invites Indian-origin scientists & technocrats (e.g., Sundar Pichai, Satya Nadella types in nano field) for conferences & collaborations.

Current Status (2018 data)

- India ranked **3rd highest** in the world in **nano technology research** (2300 research papers published).
- However, major gap exists between **lab research** and **market-oriented applications**.
- Most research is academic/funded by government → limited innovation & patents.
- Private sector participation is still limited.

Key Challenge: Convert research output into commercial products and market impact.

Summary

Nanotechnology studies materials in the **1–100 nm** size range where new properties emerge due to quantum effects and extremely high surface-to-volume ratio.

Approaches: Top-down (breaking bulk) and Bottom-up (assembling atoms/molecules). Future lies in bottom-up.

Applications: Health (targeted drugs), Environment (water purification), Energy (fuel cells), Agriculture, Electronics (graphene transistors, flexible displays), Textiles (self-cleaning, wrinkle-free, protective clothing), Defence (nano sensors, lighter armour).

Issues: Health risks (crossing blood-brain barrier), socio-ethical disparity (only rich can afford), environmental nano-pollution, high

cost, and hypothetical risks like Grey Goo/Green Goo.

India's Position: Strong in fundamental research (3rd globally), Nano Mission since 2007, INST Mohali, Nano Park in Bengaluru, ICONSAT conferences. Gap remains in converting research into marketable products. Private sector participation and innovation need to be strengthened.

Key Takeaway: Nanotechnology is a **general-purpose technology** with huge transformative potential across all sectors, but it must be developed responsibly with focus on safety, ethics, affordability, and bridging lab-to-market gap. India has made a good start in research but needs faster commercial translation.